

A Hydrodynamic Approach to Macroscopic Gravity: Thermodynamic Equilibrium via Viscous Dissipation and Quaternion Tensors

Jung Soo Kim (Independent Researcher)

ORCID: 0009-0006-5140-0604

[Notice of Integration into the Grand Unified Framework]

This manuscript serves as a foundational preliminary study. The theoretical and mathematical mechanisms presented herein have been fully integrated, refined, and established in the final 2026 “Grand Unified Framework” paper. Readers are strongly encouraged to refer to the unified framework for the complete topological resolution.

Abstract

This paper presents a novel hydrodynamic framework for macroscopic gravity, bridging the gap between Einstein’s tensor geometry and Navier-Stokes fluid dynamics. By modeling the gravitational field as a continuous flow of Vacuum Fluctuation Energy, a dimensionally calibrated quaternion tensor (q_c) is introduced to resolve the topological coordinate singularity (gimbal lock phenomenon) inherent in rotating celestial bodies. Furthermore, to prevent the non-physical gravitational collapse observed in conventional mass-density singularity models, an isotropic hydrostatic pressure tensor ($-P\mathbf{I}$) is coupled with a radial density stratification function ($\rho(\mathbf{r})$). This system reaches a permanent thermodynamic equilibrium state by suppressing the exponential divergence of turbulent kinetic energy (TKE) through a kinematic viscous dissipation term opposing the non-linear convective vortex acceleration ($\mathbf{v} \times (\nabla \times \mathbf{v}) - \nu \nabla^2 \mathbf{v}$). This hydrodynamic formulation successfully mitigates infinite density singularities without relying on the constraints of quantum gravity, providing a computable closed-form macroscopic field theory for planetary geometry.

1 Introduction

The intersection of general relativity and fluid dynamics has historically provided a fundamental understanding of macroscopic gravity [1, 2]. Traditionally, gravitational fields have been described through geometric tensors. However, when modeling the continuous dynamics of a rotating planet, conventional coordinate systems have exhibited critical mathematical limitations, such as topological coordinate singularities (gimbal lock phenomenon) and the infinite density collapse of internal mass structures.

To address this problem beyond the limits of existing theoretical models, this paper proposes a hydrodynamic framework that treats the macroscopic gravitational field as a continuous flow of Vacuum Fluctuation Energy. Moving away from conventional descriptions that rely solely on geometric curvature, we introduce a hydrodynamic perspective that inherently incorporates the physical resistance and spatial thermodynamic capacity inside celestial bodies.

By applying the proper quaternion tensor representation (q_c), the topological coordinate singularity inherent in three-dimensional rotating celestial bodies is resolved, enabling singularity-free angular momentum calculations. Furthermore, to prevent the structural implosion of a point-mass model, an isotropic hydrostatic pressure tensor ($-P\mathbf{I}$) is coupled with a radial density stratification function ($\rho(\mathbf{r})$). Finally, by integrating a kinematic viscous dissipation term to balance the convective vortex acceleration [2], the system is structured to mitigate turbulent kinetic energy and strictly maintain thermodynamic equilibrium. This formulation provides an analytically computable alternative for macroscopic gravitational field analysis.

2 Theoretical Framework and Hydrodynamic Equations

This study builds progressively from the fundamental gravitational model of classical mechanics to a hydrodynamic closed-form tensor model capable of mathematically validating the thermodynamic equilibrium of macroscopic planetary systems.

Step 1: Classical Mass-Distance Model (Newtonian Mechanics) The foundational approach to gravity begins with the classical model describing the attractive force between two point masses:

$$\mathbf{F} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}} \quad (1)$$

While this model is effective for predicting macroscopic orbits between celestial bodies, it fails to account for the physical resistance inside planets or the continuous hydrodynamic properties of the medium (spacetime).

Step 2: Geometric Curvature Model (General Relativity Basis) To reflect the flow of fluid or energy, classical mechanics evolved into a tensor form dealing with the curvature of spacetime via the energy-momentum tensor ($T_{\mu\nu}$) of Einstein's field equations [1]. By applying a macroscopic control-volume spatial contraction, this tensor is mathematically mapped onto the topological boundary ($\partial\Omega$) using the macroscopic mass-energy moment vector ($\mathbf{M}$) and an effective Energy Influx Tensor ($\mathbf{R}_X$), yielding:

$$\Phi_{\text{geo}} = \oint_{\partial\Omega} \mathbf{M} \otimes (\mathbf{R}_X \cdot \hat{\mathbf{n}}) \, d\Sigma \quad (2)$$

However, this equation induces mathematical discontinuity at the poles of a rotating celestial body, generating a singularity problem where the internal density converges to infinity.

Step 3: Introduction of Rotation and Pressure Resistance (Fluid-Structure Equilibrium) To resolve the topological coordinate singularity and the density collapse problem, this study introduces the localized Angular Momentum Flux Coefficient ($\mathcal{K}_\omega$) and the quaternion rotation tensor (q_c). By coupling them via a dyadic tensor product, a symmetric rank-2 tensor is generated. Introducing the isotropic hydrostatic pressure tensor ($-P\mathbf{I}$), we secure dimensional homogeneity (Pa) and define a stable intermediate flux:

$$\Phi_{\text{hydro}} = \oint_{\partial\Omega} \mathbf{r} \otimes \left(\left[\mathcal{K}_\omega \rho(\mathbf{r}) (q_c \omega q_c^*)^{\otimes 2} + \mathbf{R}_X - P\mathbf{I} \right] \cdot \hat{\mathbf{n}} \right) \, d\Sigma \quad (3)$$

At this stage, the planet mathematically secures the morphological stability of a non-collapsing sphere. Nevertheless, the continuous injection of turbulent kinetic energy (TKE) caused by the fluid's rotation remains an ongoing dynamic challenge.

Step 4: Viscous Dissipation and Thermodynamic Equilibrium (Final Equation) To prevent the exponential divergence of rotational energy and guide the system into a permanent geometric equilibrium, the non-linear convective vortex acceleration is merged into the macroscopic tensor space alongside the kinematic viscous dissipation term. Applying the Gauss Divergence Theorem, the macroscopic boundary surface integration is rigorously mapped to the internal volumetric entropy dissipation. Establishing a critical scalar threshold (Γ_0) where the Energy Influx ($\mathbf{R}_X$) and Efflux ($\mathbf{T}_X$) interact, the final geometrically

consistent tensor equation is formulated over the unified boundary ($\partial\Omega$) utilizing the Kakeya projection framework [4]:

$$\begin{aligned}\Phi &= \text{Re} \left\{ \oint_{\partial\Omega} \mathbf{r} \otimes \left(\left(\mathbb{P}_{\text{Kakeya}} : \left[\mathcal{K}_\omega \rho(\mathbf{r}) (q_c \boldsymbol{\omega} q_c^*)^{\otimes 2} + \Gamma_0 \mathbf{R}_X - P\mathbf{I} + \mathbf{T}_X \right] \right) \cdot \hat{\mathbf{n}} \right) d\Sigma \right\} \\ &= \mathcal{E}_{\text{ZPF}} - \int_0^t \int_{\Omega} \mathcal{R}_{\text{entropy}} \left(\mathbf{v} \times (\nabla \times \mathbf{v}) - \nu \nabla^2 \mathbf{v} \right) e^{-\lambda\tau} dV d\tau\end{aligned}\quad (4)$$

This equation upgrades the geometric gravitational model into a Navier-Stokes-based hydrodynamic thermodynamic equilibrium state, providing a framework for computing macroscopic fields while avoiding singularity errors. Here, $\partial\Omega$ represents the Lipschitz continuous topological boundary, Ω is the enclosed macroscopic control volume, $\hat{\mathbf{n}}$ is the normal vector, λ is the non-linear damping constant, τ is the time delay variable, and $\mathcal{E}_{\text{ZPF}}$ denotes the Zero-Point Fluctuation Equilibrium Tensor. $\mathcal{R}_{\text{entropy}}$ functions as the rank-3 mapping operator that translates the localized kinematic vectors into the macroscopic tensor space.

3 Simulation Methodology and Analytical Stability Results

To verify the reliability of the macroscopic gravity hydrodynamic equation proposed in this study, the major computational singularities that occur in a Computational Fluid Dynamics (CFD) environment were analyzed. It is analytically demonstrated how the mathematical defense mechanisms inherent in the formula structurally stabilize the system.

3.1 CFD Discretization and Quaternion-Based Computational Stability

When conventional Euler angle-based rotation matrices are substituted into a 3D grid simulation, the radius of curvature converges to zero at the rotational poles, inevitably causing a 'Divide by Zero' topological singularity. This computational model discretizes the angular velocity vector by mapping it to a quaternion tensor (q_c). Through this, the non-linear topological disconnection at the polar coordinates is eliminated, analytically confirming that continuous fluid computation remains computationally robust.

3.2 Implosion Defense via Hydrostatic Tensor

Point-mass-based gravity simulations inherently face a mathematical divergence limit where density and gravitational acceleration approach infinity. Within this simulation model, the radial density stratification function ($\rho(\mathbf{r})$) and the isotropic hydrostatic pressure tensor ($-P\mathbf{I}$) interact directly with the Energy Influx Tensor ($\mathbf{R}_X$). Analytical integration indicates that this pressure defense term acts as an effective buffer mitigating fluid implosion, maintaining the morphological stability of the planetary mesh.

3.3 TKE Control and Thermodynamic Equilibrium Convergence

In environments dominated by fluid vorticity ($\nabla \times \mathbf{v}$), the exponential divergence of turbulent kinetic energy (TKE) frequently induces numerical instability. This framework allocates the kinematic viscous dissipation term ($-\nu \nabla^2 \mathbf{v}$) to counteract the convective vortex acceleration ($\mathbf{v} \times (\nabla \times \mathbf{v})$) within the spatial boundary. As the computation progresses, surplus kinetic energy is irreversibly converted into thermal entropy. Consequently, the internal energy of the computational network asymptotically converges to the Zero-Point Fluctuation Energy equilibrium state ($\mathcal{E}_{\text{ZPF}}$).

4 Consideration on the Prime Topology Expansion of Macroscopic Space-time Vacuum Fluid

Extending the four-step viscous dissipation equation beyond a local planetary system into an energy equilibrium model for the macroscopic universe secures mathematical continuity across broader scales. By mapping the flow of the vacuum fluid to a specific topological network (Prime Topology) to control dimensional interference, it establishes the foundation for a singularity-free hydrodynamic model based on zero-point

equilibrium, preserving geometric limits even under extreme energy influx environments. The specific mathematical formulation of this geometric tensor expansion will be systematically elucidated through subsequent research based on this framework.

5 Conclusion and Future Directions

This study transcends the traditional perspective that interprets macroscopic gravity merely as the passive curvature of spacetime, shedding new light on it as a geometric equilibrium state where microscopic fluctuations of the Zero-Point Field manifest into a three-dimensional physical continuum [3]. By integrating Einstein's geometric tensor with Navier-Stokes-based hydrodynamic resistance, we have elucidated the mechanism by which macroscopic gravity within a closed system maintains symmetrical stability without infinite density collapse, utilizing the quaternion tensor (q_c) and the isotropic hydrostatic pressure tensor ($-P\mathbf{I}$).

However, extending this topological stability to continuous simulations at the fluid dynamic and thermodynamic levels presents significant computational challenges for existing analytical frameworks. The conventional linear viscosity model and the kinematic viscous dissipation term ($-\nu\nabla^2\mathbf{v}$) alone are insufficient to comprehensively address the nonlinear divergence generated during the turbulence of the vacuum fluid. This implies an inherent singularity limitation within standard CFD formulations.

To manage this hydrodynamic resistance and achieve permanent dynamic equilibrium, an advanced macroscopic and microscopic tensor system is required. It is essential to redefine space itself as the rotational axis of a Nonlinear Multi-scale Scalar Field where energy circulates, and to introduce a novel geometric equation incorporating the vacuum pressure gradient (∇P_{vacuum}).

Therefore, this study serves as the foundational prologue toward a larger-scale theoretical expansion that redefines the fundamental interaction between gravity and the vacuum fluid. The hydrodynamic tensor framework presented herein identifies the topological limitations of current CFD environments and highlights the necessity for an expanded equation formulation, which will be sequentially published in a forthcoming three-part paper series.

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Declarations and License

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